

Technical Specifications of Matrice M210 with Zenmuse XT2

Table 6

Operating Frequency	2.4000 to 2.4835GHz; 5.725 to 5.850GHz
Hovering Accuracy (P-mode with GPS)	- Vertical: $\pm 0.5\text{m}$ or $\pm 0.1\text{m}$, downward vision system enabled - Horizontal: $\pm 1.5\text{m}$ or $\pm 0.3\text{m}$, downward vision system enabled
Ascent Speed	5m/s maximum
Max Speed	- Dual downward gimbal/single upward gimbal: S-mode/A-mode: 73.8kph; P-mode: 61.2kph - Single downward gimbal (gimbal connector I): S-mode/A-mode: 81kph; P-mode: 61.2kph
Dimensions	- Unfolded, propellers, and landing gears included, 883mm x 886mm x 427mm - Folded, propellers, and landing gears excluded, 722mm x 282mm x 242mm
Zenmuse XT2:	
FPA & Digital Video Display Format	640 x 512
Scene Range	High gain: -13 to 275°F (-25 to 135°C) Low gain: -40 to 1022°F (-40 to 550°C)
Dimensions	118.02mm x 111.6mm x 125.5mm
Thermal Imager	Uncooled VOx microbolometer

Top agricultural drones

Drones are made keeping their application in mind. Here are the top agricultural drones for 2021:

DJI Agras MG-1 for crop spraying. The Agras MG-1S from DJI integrates cutting-edge technologies including the new A3 flight controller and a radar sensing system that provides additional reliability during flight. The spraying system and flow sensor ensure accurate operations. When used with the MG Intelligent Operation Planning System and the DJI Agriculture Management Platform, a user can plan operations, manage flights in real time, and closely monitor aircraft operation. The MG-1S is a high-performance aircraft capable of offering comprehensive solutions for agricultural care.

senseFly eBee SQ for overall agriculture use. senseFly eBee SQ is an advanced agricultural drone built around Parrot's ground-breaking Sequoia camera. This fully integrated

and highly precise multispectral sensor captures data across four non-visible bands, plus visible RGB imagery, in just one flight. With this precise data you can generate accurate index maps and use these to create high-quality prescriptions for carefully optimising crop treatments to improve production quality, increase yields, and reduce costs. The eBee SQ can cover hundreds of acres in a single flight—up to ten times more ground than quadcopter drones—for extremely efficient crop monitoring and analysis. This means fewer flights in total, far less time spent collecting data and more time acting on it.

Delair UX 11 AG for agriculture mapping. The Delair UX11 AG is a plant mapping drone, allowing you to collect land aerial intelligence more accurately and efficiently. The drone is capable of onboard data processing and, with wireless and 3G/4G communications, allowing you to overlay maps for temporal analysis. Unlike the other drones, this is a fixed-wing



Fig. 7: FLIR Matrice M210 with Zenmuse XT2

drone, which means it is capable of flying longer distances and for longer periods of time (up to 50 minutes, across 47 kilometres). The drone carries a high-end multispectral camera for plant level measures.

FLIR Matrice M210 with Zenmuse XT2 for thermal imaging. The Matrice M210 can carry large payloads, whether it's a third-party sensor and cameras from multi-spectral to hyper-spectral and full frame, or its own DJI Zenmuse XT2 dual thermal camera. The drone can be upgraded with DJI's D-RTK GNSS system to allow for centimetre level hovering accuracy. It's also compatible with DJI's SDK, so users can control up to five aircraft simultaneously, making inspecting a large farmland more efficient than previously possible. The Zenmuse XT2 is also compatible with DJI's M200 series, M200 series V2, and M600 Pro drones. **EFY**

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